

Романов Александр Дмитриевич
241-321 РРР №1 Вариант 1

Задача 1

$$A = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 3 & 2 \\ 4 & 0 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 & 2 \\ 0 & -3 & 3 \\ 1 & 1 & -1 \end{pmatrix}$$

$$A^2 + 3B^T$$

$$A^2 = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 3 & 2 \\ 4 & 0 & -3 \end{pmatrix} \cdot \begin{pmatrix} -1 & 1 & 0 \\ 1 & 3 & 2 \\ 4 & 0 & -3 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 2 \\ 10 & 10 & 0 \\ -16 & 4 & 9 \end{pmatrix}$$

$$B^T = \begin{pmatrix} 2 & 0 & 1 \\ 0 & -3 & 1 \\ 2 & 3 & -1 \end{pmatrix}; \quad 3B^T = \begin{pmatrix} 6 & 0 & 3 \\ 0 & -9 & 3 \\ 6 & 9 & -3 \end{pmatrix}$$

$$A^2 + 3B^T = \begin{pmatrix} 2 & 2 & 2 \\ 10 & 10 & 0 \\ -16 & 4 & 9 \end{pmatrix} + \begin{pmatrix} 6 & 0 & 3 \\ 0 & -9 & 3 \\ 6 & 9 & -3 \end{pmatrix} = \begin{pmatrix} 8 & 2 & 5 \\ 10 & 1 & 3 \\ -10 & 13 & 6 \end{pmatrix}$$

Задача 3



$$\begin{vmatrix} 1 & 0 & -1 & 2 \\ 0 & 2 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 2 & 1 \end{vmatrix} \xrightarrow{\substack{\text{III}-\text{I} \\ \text{IV}-\text{I}}} \begin{vmatrix} 1 & 0 & -1 & 2 \\ 0 & 2 & 1 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & -1 & -1 \end{vmatrix} = \\
 = 1 \cdot \begin{vmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & -1 \end{vmatrix} = 2 \cdot (-1) - 1 \cdot (-1) - 0 - 2 + 1 = -2 - 1 = -3$$

Задача 4

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{pmatrix}; |A| = 3 + 0 + 4 + 2 - 0 - 0 = 9 \neq 0, \quad A^{-1} \exists$$

$$A^T = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 2 \\ 2 & 0 & 3 \end{pmatrix}; \tilde{A} = \begin{pmatrix} 3 & 4 & -2 \\ -3 & 5 & 2 \\ 3 & -2 & 1 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} \frac{3}{9} & \frac{4}{9} & \frac{-2}{9} \\ \frac{-3}{9} & \frac{5}{9} & \frac{2}{9} \\ \frac{3}{9} & \frac{-2}{9} & \frac{1}{9} \end{pmatrix}$$

Проверим:

$$A \cdot A^{-1} = \downarrow$$

$$A \cdot A^{-1} = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} \frac{3}{9} & \frac{4}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{5}{9} & \frac{2}{9} \\ \frac{3}{9} & \frac{2}{9} & \frac{1}{9} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = E$$

Задача 5

$$A X B = C$$

$$X = A^{-1} C B^{-1}$$

$$A = \begin{pmatrix} 2 & 6 \\ -1 & -1 \end{pmatrix}; |A| = -2 + 6 = 4 \neq 0, A^{-1} \exists$$

$$A^{-1} = \begin{pmatrix} \frac{-1}{4} & \frac{-6}{4} \\ \frac{1}{4} & \frac{2}{4} \end{pmatrix}$$

$$B = \begin{pmatrix} 2 & 3 \\ 5 & 1 \end{pmatrix}; |B| = 2 - 15 = -13 \neq 0, B^{-1} \exists$$

$$B^{-1} = \begin{pmatrix} \frac{-1}{13} & \frac{3}{13} \\ \frac{5}{13} & \frac{-2}{13} \end{pmatrix}$$

$$C = \begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix}$$

$$X = \begin{pmatrix} \frac{-1}{4} & \frac{-6}{4} \\ \frac{1}{4} & \frac{2}{4} \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix} \cdot \begin{pmatrix} \frac{-1}{13} & \frac{3}{13} \\ \frac{5}{13} & \frac{-2}{13} \end{pmatrix} = \begin{pmatrix} 1 & -6 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix} \cdot \begin{pmatrix} 2 & -12 \\ -1 & 4 \end{pmatrix} \cdot \begin{pmatrix} \frac{-1}{13} & \frac{3}{13} \\ \frac{5}{13} & \frac{-2}{13} \end{pmatrix} = \begin{pmatrix} \frac{-31}{13} & \frac{-51}{13} \\ \frac{10}{13} & \frac{-16}{13} \end{pmatrix}$$

Задача 6

$$\begin{array}{c}
 \left(\begin{array}{ccccc} 1 & -1 & 2 & 3 & 7 \\ 0 & 5 & 4 & 3 & 1 \\ 2 & 3 & 8 & 9 & 15 \\ 3 & -8 & 2 & 6 & 20 \end{array} \right) \begin{array}{l} \\ \\ \text{III} - 2\text{I} \\ \text{IV} - 3\text{I} \end{array} \sim \left(\begin{array}{ccccc} 1 & -1 & 2 & 3 & 7 \\ 0 & 5 & 4 & 3 & 1 \\ 0 & 5 & 4 & 3 & 1 \\ 0 & -5 & -4 & -3 & -1 \end{array} \right) \begin{array}{l} \\ \\ \text{III} - \text{II} \\ \text{IV} + \text{II} \end{array} \\
 \sim \left(\begin{array}{ccccc} 1 & -1 & 2 & 3 & 7 \\ 0 & 5 & 4 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \begin{array}{l} 1 \\ 2 \\ \\ \end{array} \quad r(A) = 2
 \end{array}$$

Задача 7

$$\begin{cases} x_1 + 4x_2 - x_3 = -2 \\ 2x_1 + 2x_2 + 3x_3 = 7 \\ 3x_1 + 6x_2 - 2x_3 = 1 \end{cases}$$

$$A = \begin{pmatrix} 1 & 4 & -1 \\ 2 & 2 & 3 \\ 3 & 6 & -2 \end{pmatrix}; \quad B = \begin{pmatrix} -2 \\ 7 \\ 1 \end{pmatrix}$$

$$|A| = -4 + 36 - 12 + 6 + 16 - 12 = 24 \neq 0, \text{ ф.к.р. нр-нр.}$$

$$\Delta_1 = \begin{vmatrix} -2 & 4 & -1 \\ 7 & 2 & 3 \\ 1 & 6 & -2 \end{vmatrix} = 8 + 12 - 42 + 2 + 56 + 36 = 72$$

$$X_1 = \frac{\Delta_1}{|A|} = \frac{72}{24} = 3$$

$$\Delta_2 = \begin{vmatrix} 1 & -2 & -1 \\ 2 & 7 & 3 \\ 3 & 1 & -2 \end{vmatrix} = -14 - 18 - 2 + 21 - 6 - 3 \Rightarrow -24$$

$$X_2 = \frac{-24}{24} = -1$$

$$\Delta_3 = \begin{vmatrix} 1 & 4 & -2 \\ 2 & 2 & 7 \\ 3 & 6 & 1 \end{vmatrix} = 2 + 84 - 24 + 12 - 8 - 42 \Rightarrow 24$$

$$X_3 = \frac{24}{24} = 1$$

Ответ: $(3; -1; 1)$

Проверка:

$$3 + 4 - 3 - 4 - 1 = -2$$

$$-2 = -2 \quad \checkmark$$

Задача 8

$$A = \begin{pmatrix} 1 & 4 & -1 \\ 2 & 2 & 3 \\ 3 & 6 & -2 \end{pmatrix}; X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}; B = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} -2 \\ 7 \\ 1 \end{pmatrix}$$



$$A \cdot X = B$$

$$X = A^{-1} \cdot B$$

$$|A| = -4 + 36 + 6 - 12 + 16 - 18 = 24 \neq 0, A^{-1}$$

$$A^T = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 2 & 6 \\ -1 & 3 & -2 \end{pmatrix}; \quad A^{-1} = \begin{pmatrix} -22 & 2 & 14 \\ 13 & 1 & -5 \\ 6 & 6 & -6 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} \frac{-22}{24} & \frac{2}{24} & \frac{14}{24} \\ \frac{13}{24} & \frac{1}{24} & \frac{-5}{24} \\ \frac{6}{24} & \frac{6}{24} & \frac{-6}{24} \end{pmatrix}$$

$$X = \begin{pmatrix} \frac{-22}{24} & \frac{2}{24} & \frac{14}{24} \\ \frac{13}{24} & \frac{1}{24} & \frac{-5}{24} \\ \frac{6}{24} & \frac{6}{24} & \frac{-6}{24} \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 7 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

Answer: (3; -1; 1)

Задание 9



$$\begin{cases} X_1 + 4X_2 - X_3 = -2 \\ 2X_1 + 2X_2 + 3X_3 = 7 \\ 3X_1 + 6X_2 - 2X_3 = 1 \end{cases}$$

$$A|B = \left(\begin{array}{ccc|c} 1 & 4 & -1 & -2 \\ 2 & 2 & 3 & 7 \\ 3 & 6 & -2 & 1 \end{array} \right) \xrightarrow[\text{III}-3\text{I}]{\text{II}-2\text{I}} \left(\begin{array}{ccc|c} 1 & 4 & -1 & -2 \\ 0 & -6 & 5 & 11 \\ 0 & -6 & 1 & 7 \end{array} \right) \xrightarrow{\text{III}-\text{II}}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 4 & -1 & -2 \\ 0 & -6 & 5 & 11 \\ 0 & 0 & -4 & -4 \end{array} \right) \begin{matrix} 1 \\ 2 \\ 3 \end{matrix}$$

$r(A) = r(A|B) = 3$ — матрица совестна

$r = n = 3$ — матрица определённая

$$\begin{cases} X_1 + 4X_2 - X_3 = -2 \\ 0 - 6X_2 + 5X_3 = 11 \\ 0 + 0 - 4X_3 = -4 \end{cases} \Rightarrow X_3 = 1$$

$$\begin{cases} X_1 + 4X_2 = -1 \\ -6X_2 = 6 \end{cases} \Rightarrow X_2 = -1$$

$$X_1 = 3$$

Ответ: $(3; -1; 1)$

Задача 10

$$\begin{array}{c} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 5 \\ 3 & 2 & -1 & -2 & 1 \\ 5 & 4 & 0 & 4 & 11 \end{array} \right) \xrightarrow[\text{III} - 5\text{I}]{\text{II} - 3\text{I}} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 5 \\ 0 & -1 & -10 & -11 & -14 \\ 0 & -1 & -10 & -11 & -14 \end{array} \right) \sim \\ \xrightarrow{\text{II} - \text{I}} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 5 \\ 0 & -1 & -10 & -11 & -14 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \end{array}$$

$\begin{matrix} x_1 & x_2 & t_1 & t_2 \\ \hline 1 & 1 & 2 & 3 \\ 0 & -1 & -10 & -11 \\ 0 & 0 & 0 & 0 \end{matrix}$

$r(A) = r(A/B) = 2$ — система совместна

$r < n$ — система неопределима

$$\Delta = \begin{vmatrix} 1 & 1 \\ 0 & -1 \end{vmatrix} = -1 \neq 0$$

Пусть $x_2 = t_1$, $x_4 = t_2$.

Тогда

$$\begin{cases} x_1 + x_2 + 2t_1 + 3t_2 = 5 \\ -x_2 - 10t_1 - 11t_2 = -14 \end{cases} \Rightarrow x_2 = 14 - 10t_1 - 11t_2$$

$$x_1 + 14 - 10t_1 + 2t_1 - 11t_2 + 3t_2 = 5$$

$$x_1 = -9 + 8t_1 + 8t_2$$

Ответ: система совместна, неопределима.

$$\text{o.p. } (-9+8t_1+8t_2; 14-10t_1-11t_2; t_1; t_2)$$

$$\text{z.p. } (7; -7; 1; 1)$$

Задача 11

$$\begin{cases} 2x_1 + 6x_2 - 9x_3 - 9x_4 = 0 \\ 4x_1 - 5x_2 + x_4 = 0 \\ 3x_1 + 3x_2 - 7x_3 - 4x_4 = 0 \\ x_1 - 3x_2 + 2x_3 + 5x_4 = 0 \end{cases}$$

$$\begin{aligned} & \left(\begin{array}{cccc|c} 2 & 6 & -9 & -9 & 0 \\ 4 & 0 & -5 & 1 & 0 \\ 3 & 3 & -7 & -4 & 0 \\ 1 & -3 & 2 & 5 & 0 \end{array} \right) \xrightarrow{\text{I}-\text{IV}} \left(\begin{array}{cccc|c} 1 & 9 & -11 & -14 & 0 \\ 4 & 0 & -5 & 1 & 0 \\ 3 & 3 & -7 & -4 & 0 \\ 1 & -3 & 2 & 5 & 0 \end{array} \right) \xrightarrow{\begin{array}{l} \text{II}-4\text{I} \\ \text{III}-3\text{I} \\ \text{IV}-\text{I} \end{array}} \\ & \sim \left(\begin{array}{cccc|c} 1 & 9 & -11 & -14 & 0 \\ 0 & -36 & 39 & 57 & 0 \\ 0 & -24 & 26 & 38 & 0 \\ 0 & -12 & 13 & 19 & 0 \end{array} \right) \xrightarrow{\begin{array}{l} \text{II}-2\text{IV} \\ \text{III}-2\text{IV} \end{array}} \left(\begin{array}{cccc|c} 1 & 9 & -11 & -14 & 0 \\ 0 & -12 & 13 & 19 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -12 & 13 & 19 & 0 \end{array} \right) \xrightarrow{\text{IV}-\text{II}} \\ & \sim \left(\begin{array}{cccc|c} 1 & 9 & -11 & -14 & 0 \\ 0 & -12 & 13 & 19 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad \begin{array}{l} \downarrow L_3 \\ \downarrow L_4 \end{array} \end{aligned}$$

Пусто $x_2 = L_3; x_4 = L_4$
Тогда:



$$\begin{cases} X_1 + 9X_2 - 11L_3 - 14L_4 = 0 \\ -12X_2 + 13L_3 + 19L_4 = 0 \end{cases} \Rightarrow X_2 = \frac{13L_3 + 19L_4}{12}$$

$$X_1 + \frac{9 \cdot 13L_3}{12} + \frac{9 \cdot 19L_4}{12} - 11L_3 - 14L_4 = 0$$

$$X_1 = 11L_3 - 9\frac{3}{4}L_3 + 14L_4 - 14\frac{1}{4}L_4 = \frac{1}{4}L_3 + \frac{1}{4}L_4$$

Donc la p.e.m.

$$X_1 = \frac{5}{4}L_3 - \frac{1}{4}L_4$$

$$X_2 = \frac{13}{12}L_3 + \frac{19}{12}L_4$$

$$X_3 = 1L_3 + 0 \cdot L_4$$

$$X_4 = 0 \cdot L_3 + 1 \cdot L_4$$

P.C.P.

$$\begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{pmatrix} = L_3 \begin{pmatrix} \frac{5}{4} \\ \frac{13}{12} \\ 1 \\ 0 \end{pmatrix} + L_4 \begin{pmatrix} -\frac{1}{4} \\ \frac{19}{12} \\ 0 \\ 1 \end{pmatrix}$$